

IRRIGATION, INPUTS, CREDIT, INSURANCE AND SUSTAINABLE AGRICULTURE

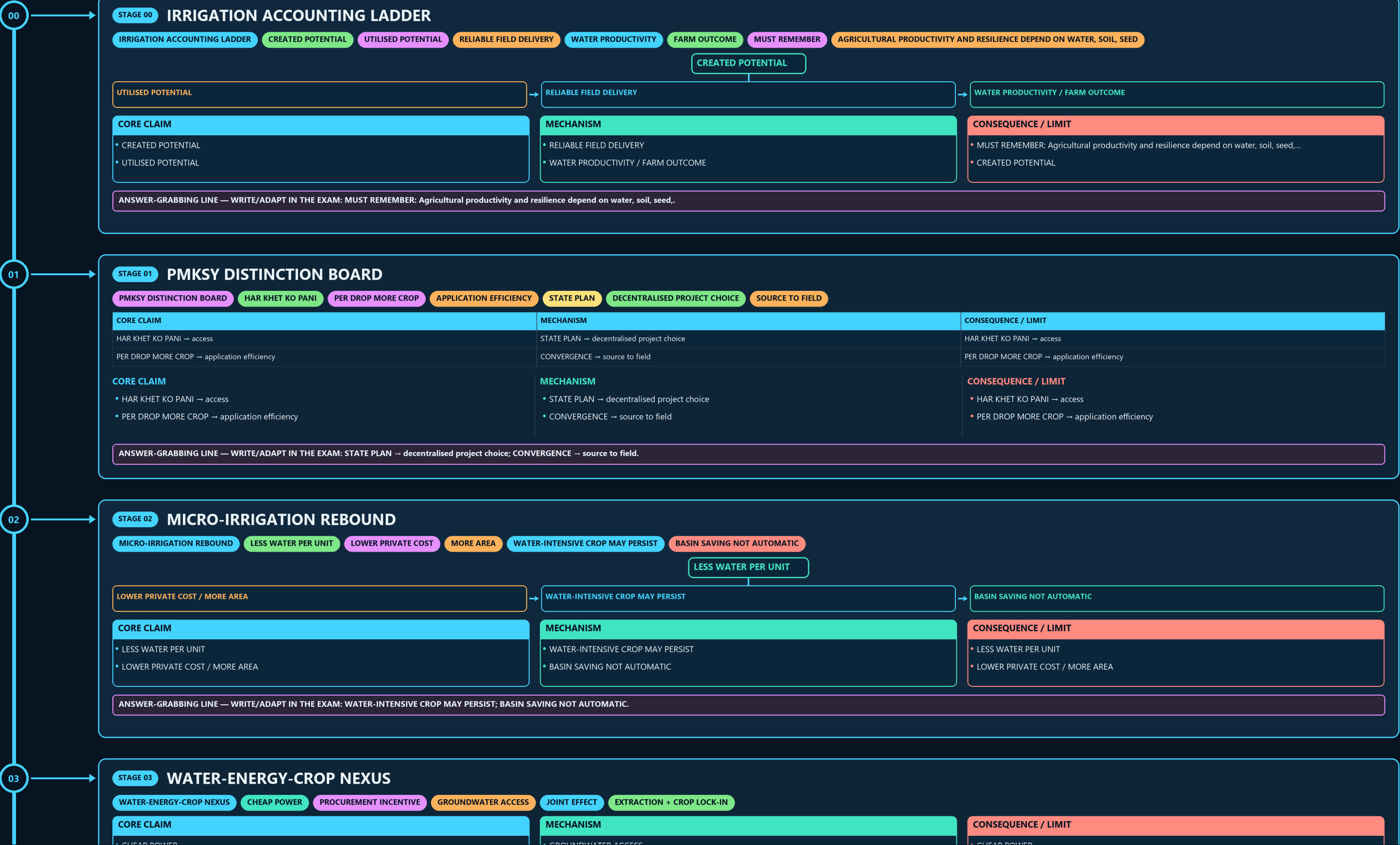
IRRIGATION ACCOUNTING LADDER → PMKSY DISTINCTION BOARD → MICRO-IRRIGATION REBOUND → WATER-ENERGY-CROP NEXUS → WATERSHED RAIL → GROUNDWATER GOVERNANCE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



LOWER PRIVATE COST / MORE AREA

→

WATER-INTENSIVE CROP MAY PERSIST

→

BASIN SAVING NOT AUTOMATIC

CORE CLAIM

- LESS WATER PER UNIT
- LOWER PRIVATE COST / MORE AREA

MECHANISM

- WATER-INTENSIVE CROP MAY PERSIST
- BASIN SAVING NOT AUTOMATIC

CONSEQUENCE / LIMIT

- LESS WATER PER UNIT
- LOWER PRIVATE COST / MORE AREA

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER-INTENSIVE CROP MAY PERSIST; BASIN SAVING NOT AUTOMATIC.

03

STAGE 03

WATER-ENERGY-CROP NEXUS

WATER-ENERGY-CROP NEXUS

CHEAP POWER

PROCUREMENT INCENTIVE

GROUNDWATER ACCESS

JOINT EFFECT

EXTRACTION + CROP LOCK-IN

CORE CLAIM

- CHEAP POWER
- PROCUREMENT INCENTIVE

MECHANISM

- GROUNDWATER ACCESS
- JOINT EFFECT → extraction + crop lock-in

CONSEQUENCE / LIMIT

- CHEAP POWER
- PROCUREMENT INCENTIVE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GROUNDWATER ACCESS; JOINT EFFECT → extraction + crop lock-in.

04

STAGE 04

WATERSHED RAIL

WATERSHED RAIL

VEGETATION + CONTOUR TREATMENT

BUNDS + MOISTURE RETENTION

RECHARGE STRUCTURES

PRODUCTIVE USE + MAINTENANCE

RIDGE → vegetation + contour treatment

→

MID-SLOPE → bunds + moisture retention

→

DRAINAGE → check / recharge structures

→

VALLEY → productive use + maintenance

CORE CLAIM

- RIDGE → vegetation + contour treatment
- MID-SLOPE → bunds + moisture retention

MECHANISM

- DRAINAGE → check / recharge structures
- VALLEY → productive use + maintenance

CONSEQUENCE / LIMIT

- RIDGE → vegetation + contour treatment
- MID-SLOPE → bunds + moisture retention

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DRAINAGE → check / recharge structures; VALLEY → productive use + maintenance.

05

STAGE 05

GROUNDWATER GOVERNANCE

GROUNDWATER GOVERNANCE

PRIVATE PUMP

SHARED AQUIFER

LOCAL WATER BUDGET

CROP + EXTRACTION RULES

CLOSE DISTINCTION

IRRIGATION POTENTIAL IS NOT UTILISED IRRIGATION, CREDIT SANCTION IS

SHARED AQUIFER

→

LOCAL WATER BUDGET

→

CROP + EXTRACTION RULES

PRIVATE PUMP

↓

LOCAL WATER BUDGET

CORE CLAIM

- PRIVATE PUMP
- SHARED AQUIFER

MECHANISM

- LOCAL WATER BUDGET
- CROP + EXTRACTION RULES

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Irrigation potential is not utilised irrigation, credit sanction is...
- PRIVATE PUMP

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Irrigation potential is not utilised irrigation, credit sanction is.

06

STAGE 06

CREDIT-INSURANCE FORK

CREDIT-INSURANCE FORK

PRE-HARVEST LIQUIDITY

COVERED-LOSS TRANSFER

FINANCE ARCHITECTURE

TRIGGER + ASSESSMENT + SETTLEMENT

CORE CLAIM

KCC → pre-harvest liquidity

MECHANISM

NABARD / BANK → finance architecture

CONSEQUENCE / LIMIT

KCC → pre-harvest liquidity

CORE CLAIM

INSURANCE → covered-loss transfer

MECHANISM

CLAIM → trigger + assessment + settlement

CONSEQUENCE / LIMIT

INSURANCE → covered-loss transfer

CORE CLAIM

- KCC → pre-harvest liquidity
- INSURANCE → covered-loss transfer

MECHANISM

- NABARD / BANK → finance architecture
- CLAIM → trigger + assessment + settlement

CONSEQUENCE / LIMIT

- KCC → pre-harvest liquidity
- INSURANCE → covered-loss transfer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Irrigation potential is not utilised irrigation, credit sanction is.

06

STAGE 06 CREDIT-INSURANCE FORK

CREDIT-INSURANCE FORK PRE-HARVEST LIQUIDITY COVERED-LOSS TRANSFER FINANCE ARCHITECTURE TRIGGER + ASSESSMENT + SETTLEMENT

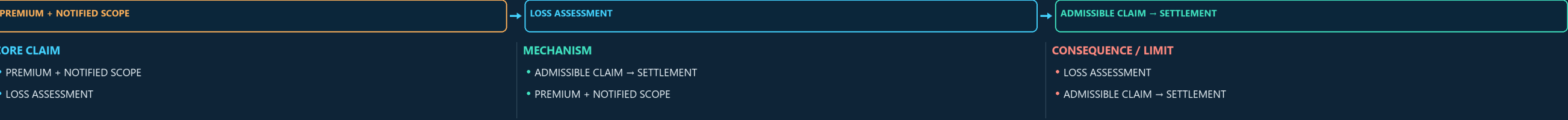
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
KCC → pre-harvest liquidity	NABARD / BANK → finance architecture	KCC → pre-harvest liquidity
INSURANCE → covered-loss transfer	CLAIM → trigger + assessment + settlement	INSURANCE → covered-loss transfer
CORE CLAIM • KCC → pre-harvest liquidity • INSURANCE → covered-loss transfer	MECHANISM • NABARD / BANK → finance architecture • CLAIM → trigger + assessment + settlement	CONSEQUENCE / LIMIT • KCC → pre-harvest liquidity • INSURANCE → covered-loss transfer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NABARD / BANK → finance architecture; CLAIM → trigger + assessment + settlement.

07

STAGE 07 INSURANCE OUTCOME CHAIN

INSURANCE OUTCOME CHAIN PREMIUM + NOTIFIED SCOPE LOSS ASSESSMENT ADMISSIBLE CLAIM



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PREMIUM + NOTIFIED SCOPE; LOSS ASSESSMENT; ADMISSIBLE CLAIM → SETTLEMENT.

08

STAGE 08 BASIS-RISK BOARD

BASIS-RISK BOARD AREA YIELD LOWER VERIFICATION COST FARM LOSS MAY DIVERGE WEATHER INDEX PROXY TRIGGER DATA + TIMELINESS DECIDE TRUST

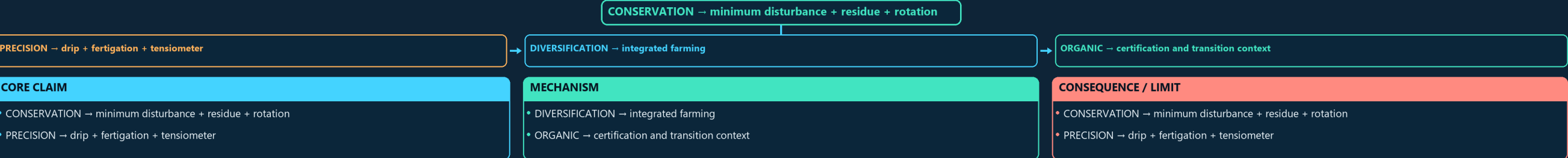
CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
AREA YIELD → lower verification cost	WEATHER INDEX → proxy trigger	AREA YIELD → lower verification cost
FARM LOSS → may diverge	DATA + TIMELINESS decide trust	FARM LOSS → may diverge
CORE CLAIM • AREA YIELD → lower verification cost • FARM LOSS → may diverge	MECHANISM • WEATHER INDEX → proxy trigger • DATA + TIMELINESS decide trust	CONSEQUENCE / LIMIT • AREA YIELD → lower verification cost • FARM LOSS → may diverge

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WEATHER INDEX → proxy trigger; DATA + TIMELINESS decide trust.

09

STAGE 09 SUSTAINABLE-PRACTICE MAP

SUSTAINABLE-PRACTICE MAP MINIMUM DISTURBANCE + RESIDUE + ROTATION DRIP + FERTIGATION + TENSIO METER INTEGRATED FARMING CERTIFICATION AND TRANSITION CONTEXT



ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONSERVATION → minimum disturbance + residue + rotation.

09

STAGE 09 SUSTAINABLE-PRACTICE MAP

SUSTAINABLE-PRACTICE MAP

MINIMUM DISTURBANCE + RESIDUE + ROTATION

DRIP + FERTIGATION + TENSIO METER

INTEGRATED FARMING

CERTIFICATION AND TRANSITION CONTEXT

CONSERVATION → minimum disturbance + residue + rotation

PRECISION → drip + fertigation + tensiometer

DIVERSIFICATION → integrated farming

ORGANIC → certification and transition context

CORE CLAIM

- CONSERVATION → minimum disturbance + residue + rotation
- PRECISION → drip + fertigation + tensiometer

MECHANISM

- DIVERSIFICATION → integrated farming
- ORGANIC → certification and transition context

CONSEQUENCE / LIMIT

- CONSERVATION → minimum disturbance + residue + rotation
- PRECISION → drip + fertigation + tensiometer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONSERVATION → minimum disturbance + residue + rotation.

10

STAGE 10 TENANT-ACCESS GATE

TENANT-ACCESS GATE

ACTUAL CULTIVATOR

LEASE EVIDENCE

INSURANCE ELIGIBILITY

COVERAGE OR EXCLUSION

ACTUAL CULTIVATOR

RECORD / LEASE EVIDENCE

CREDIT / INSURANCE ELIGIBILITY

COVERAGE OR EXCLUSION

CORE CLAIM

- ACTUAL CULTIVATOR
- RECORD / LEASE EVIDENCE

MECHANISM

- CREDIT / INSURANCE ELIGIBILITY
- COVERAGE OR EXCLUSION

CONSEQUENCE / LIMIT

- ACTUAL CULTIVATOR
- RECORD / LEASE EVIDENCE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECORD / LEASE EVIDENCE; CREDIT / INSURANCE ELIGIBILITY; COVERAGE OR EXCLUSION.

11

STAGE 11 RESILIENCE ANSWER SPINE

RESILIENCE ANSWER SPINE

SEPARATE POTENTIAL, USE AND EFFICIENCY

LINK WATER, ENERGY, CROP AND CREDIT

TRACE ENROLMENT TO CLAIM SETTLEMENT

CONCLUDE WITH AQUIFER + TENANT GOVERNANCE

EVIDENCE LIMIT

STATE UNIT, SEASON, BENEFICIARY DENOMINATOR

SEPARATE potential, use and efficiency

LINK water, energy, crop and credit

TRACE enrolment to claim settlement

CONCLUDE with aquifer + tenant governance

EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State unit, season, beneficiary denominator

CORE CLAIM

- SEPARATE potential, use and efficiency
- LINK water, energy, crop and credit

MECHANISM

- TRACE enrolment to claim settlement
- CONCLUDE with aquifer + tenant governance

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State unit, season, beneficiary denominator...
- SEPARATE potential, use and efficiency

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: FORMULA / STATUS / CAUSATION: State unit, season, beneficiary denominator.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

WATER PRICING, ELECTRICITY PRICING, PROCUREMENT INCENTIVES AND CROP CHOICE

MICRO-IRRIGATION SAVES WATER AT PLOT LEVEL, BUT BASIN-LEVEL SAVINGS

FORMAL TENANTS MAY GAIN CREDIT AND INSURANCE ACCESS, YET

AREA-YIELD INSURANCE REDUCES INDIVIDUAL VERIFICATION COSTS BUT INTRODUCES BASIS

CLIMATE ADAPTATION REQUIRES STRESS-TOLERANT SEEDS, ADVISORIES, WATER MANAGEMENT

PUBLIC EXTENSION IS A KNOWLEDGE INFRASTRUCTURE CONNECTING RESEARCH WITH

CENTRAL AND STATE IRRIGATION DEPARTMENTS

OPTIONAL SOURCE DEPTH

- CAUTION: Water pricing, electricity pricing, procurement incentives and crop choice form one
- CAUTION: Micro-irrigation saves water at plot level, but basin-level savings require limits on
- CAUTION: Formal tenants may gain credit and insurance access, yet insecure records create
- CAUTION: Area-yield insurance reduces individual verification costs but introduces basis risk.

ADVANCED DEBATE

- CAUTION: Climate adaptation requires stress-tolerant seeds, advisories, water management,
- CAUTION: Public extension is a knowledge infrastructure connecting research with local decision
- Central and state irrigation departments: build and manage major, medium and minor
- NABARD, banks and co-operatives: finance agricultural investment and seasonal

LEGACY / NUANCE

- Agricultural Insurance Company and empanelled insurers: implement crop-insurance
- ICAR, KVKs and state extension systems: provide research, advisories and farm-level
- CAUTION: Drip irrigation can reduce water applied per crop, but total aquifer extraction may
- CAUTION: Free power lowers the pumping cost seen by an individual farmer while accelerating

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.